

Blake B. Gaines

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Education: University of Connecticut

Bachelor of Science: Computer Science

Cumulative GPA: 3.99, Concentration in Computational Data Analytics

2018 – 2022

- Honors Program, Scholarship for Academic Excellence, Babbidge Scholar (2020-2022)

Bachelor of Arts: Mathematics

Cumulative GPA: 3.99, Minor in Linguistics

2018 – 2022

In Progress: PhD in Computer Science (Expected graduation in May 2027)

Cumulative GPA: 4.00, Supervised by Prof. Jinbo Bi

2022 – Present

- First place – School of Engineering's Graduate Research Poster Competition (2023)
- Department Predoctoral Fellowship (2024, 2025)

Work Experience

UConn Laboratory of ML & Health Informatics

Research assistant developing novel ML techniques for applications in bioinformatics

5/20 – Present

The Travelers Companies

Graph databases, representation learning, and LLMs for organizing internal documents

9/24 – 8/25

Boehringer Ingelheim

Developed an LLM-based question-answering pipeline for pharmaceutical researchers

8/21 – 7/22

Selected Publications

Gaines, B. B. & Bi, J. (2026). “**Characterizing the Discrete Geometry of ReLU Networks.**” International Conference on Learning Representations (ICLR). *Accepted (Oral)*: openreview.net/forum?id=TgLW2DiRDG.

Gaines, B. B., Zhu, C., & Bi, J. (2025). “**Explaining Graph Neural Networks with Mixed-Integer Programming.**” Neurocomputing, 130214. DOI: 10.1016/j.neucom.2025.130214.

Gaines, B. B., Song, M., & Bi, J. (2023). “**Mining Large-Scale Knowledge Graphs for Chemical Reaction Fingerprints.**” 2023 IEEE International Conference on Big Data DOI: 10.1109/BigData59044.2023.10386300.

Gao, Z., Wang, X., Gaines, B. B., Shi, X., Bi, J., & Song, M. (2023). “**Fragment-based deep molecular generation using hierarchical chemical graph representation and multi-resolution graph variational autoencoder.**” Molecular Informatics, 42(5), 2200215. DOI: 10.1002/minf.202200215.

Liu, Q., Wang, P.-S., Zhu, C., Gaines, B. B., Zhu, T., Bi, J., & Song, M. (2021). “**OctSurf: Efficient hierarchical voxel-based molecular surface representation for protein-ligand affinity prediction.**” Journal of Molecular Graphics and Modelling, 105, 107865. DOI: 10.1016/j.jmgm.2021.107865.

Recent Outreach & Projects

- **Relucent** (2026) – Open-source Python library for analyzing ReLU neural networks (pypi.org/project/relucent)
- **UConn AI Summer Camp** (2025) – Primary Organizer (healthinfo.lab.uconn.edu/aisummercamp)
- **Tolland Middle School** (2023–2025) – Volunteer Computer Science Instructor

Skills and Languages

Skills: Machine Learning, Natural Language Processing, Distributed Computing, Python (PyTorch, PyG, TensorFlow, NumPy, Scikit-Learn, NLTK), Cypher Query Language (Neo4j), Gurobi, Git, RDKit, ROS

Languages: English (Native), Mandarin Chinese (Professional Working Proficiency)